# IGCSE 0607 Practice Questions — E2.8 — Combined Questions & Markscheme

## Proportion

**Total Questions:** 18 — **Total Marks:** 106 Marks

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- **E2.8** Proportion
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*


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**1** $y$ is directly proportional to the square of $x$.
When $x = 3$, $y = 36$.

**(a)** Find an equation connecting $y$ and $x$.

$y =$ ..... **[2]**

**(b)** Find the value of $y$ when $x = 5$.

$y =$ ..... **[1]**

**2** $y$ is directly proportional to $(x + 2)$.
When $x = 3$, $y = 20$.

Find the value of $y$ when $x = 7$.

$y =$ ..... **[3]**

**3** $y$ is directly proportional to $x$.
When $x = 5$, $y = 15$.

**(a)** Find an equation connecting $y$ and $x$.

..... **[2]**

**(b)** Work out the value of $y$ when $x = 12$.

$y =$ ..... **[2]**

**4** $y$ is directly proportional to $(x+2)^3$.
When $x = 1$, $y = 135$.

**(a)** Find an equation for $y$ in terms of $x$.

$y =$ ..... **[2]**

**(b)** Find the value of $y$ when $x = 0$.

$y =$ ..... **[1]**

**(c)** Find the value of $x$ when $y = 320$.

$x =$ ..... **[2]**

**5** The variable $y$ is directly proportional to the square of $x$.
When $x = 3$, $y = 36$.

**(a)** Find an equation connecting $y$ and $x$.

$y =$ ..... **[2]**

**(b)** Find the value of $y$ when $x = 5$.

$y =$ ..... **[1]**

**(c)** Find the values of $x$ when $y = 196$.

$x =$ ..... **[2]**

**6** $y$ is directly proportional to the cube of $x$.
When $x = 2$, $y = 24$.

**(a)** Find an equation connecting $y$ and $x$.

$y =$ ..... **[2]**

**(b)** Work out the value of $y$ when $x = 5$.

$y =$ ..... **[1]**

**(c)** Work out the value of $x$ when $y = 81$.

$x =$ ..... **[2]**

**7** The table shows some corresponding positive values of $p$ and $q$.

| $p$ | 2 | 4 | 10 |
|---|---|---|---|
| $q$ | 100 | 25 | 4 |

**(a)** Determine the algebraic equation connecting $q$ and $p$.

$q =$ ..... **[2]**

**(b)** Find the value of $q$ when $p = 5$.

$q =$ ..... **[1]**

**(c)** Find the value of $p$ when $q = 1$.

$p =$ ..... **[2]**

**8** $P$ is directly proportional to the square root of $Q$.
$Q$ is inversely proportional to $R^3$.
When $Q = 9$, $P = 12$.
When $R = 2$, $Q = 2$.

**(a)** Find an equation for $P$ in terms of $R$.

$P =$ ..... **[5]**

**(b)** Calculate the value of $P$ when $R = 4$.

$P =$ ..... **[2]**

**9** $y$ is inversely proportional to the square of $x$.
When $x = 4$, $y = 3$.

**(a)** Find an equation connecting $y$ and $x$.

..... **[3]**

**(b)** Work out the possible values of $x$ when $y = 0.75$.

$x =$ ..... **[3]**

**10** $P$ is inversely proportional to the square of $Q$.
When $Q = 4$, $P = 1.5$.

**(a)** Find an equation connecting $P$ and $Q$.

$P =$ ..... **[3]**

**(b)** Find the value of $P$ when $Q = 10$.

$P =$ ..... **[1]**

**(c)** Find the positive value of $Q$ when $P = 24$.

$Q =$ ..... **[2]**

**11** The quantity $P$ is inversely proportional to the square root of $Q$.
When $Q = 9$, $P = 12$.

**(a) (i)** Find the value of $P$ when $Q = 144$.

$P =$ ..... **[3]**

**(ii)** Find the value of $Q$ when $P = 1.5$.

$Q =$ ..... **[2]**

**(b)** Calculate the percentage decrease in $P$ when the value of $Q$ is multiplied by 4.

..... % **[2]**

**12** $M$ is inversely proportional to the square root of $N$.
When $N = 16$, $M = 15$.

**(a)** Find an equation connecting $M$ and $N$.

$M =$ ..... **[2]**

**(b)** Find the value of $M$ when $N = 100$.

$M =$ ..... **[2]**

**(c)** Find the value of $N$ when $M = 1.2$.

$N =$ ..... **[3]**

**13** $P$ is inversely proportional to the square root of $Q$.
When $Q = 36$, $P = 4$.

**(a)** Find an equation for $P$ in terms of $Q$.

$P =$ ..... **[2]**

**(b)** Calculate the value of $P$ when $Q = 144$.

$P =$ ..... **[1]**

**(c)** Calculate the value of $Q$ when $P = 1.5$.

$Q =$ ..... **[2]**

**(d)** The value of $Q$ is increased by 300%.
Calculate the percentage decrease in the value of $P$.

..... % **[3]**

**14** **(a)** Some corresponding values of $x$ and $y$ are given below.

$$x = 2,\ y = 75 \qquad x = 4,\ y = 18.75 \qquad x = 10,\ y = 3$$

Identify which of the following models best fits the data:

$$y \propto x \qquad y \propto x^2 \qquad y \propto \frac{1}{x} \qquad y \propto \frac{1}{x^2}$$

Hence, find the equation for $y$ in terms of $x$.

$y =$ ..... **[3]**

**(b)** $P$ is directly proportional to the cube of $Q$.
When $Q$ is increased by 50%, work out the percentage increase in $P$.

..... % **[3]**

**15** **(a)** $y$ is directly proportional to the cube of $x$.

When $x$ increases by 20%, find the percentage increase in $y$.

..... % **[3]**

**(b)** The table shows some corresponding values of $v$ and $A$.

Determine the best variation model connecting $A$ and $v$ from the following list:

$$A \propto v \qquad A \propto v^2 \qquad A \propto \sqrt{v} \qquad A \propto \frac{1}{v}$$

You must show working to justify your decision.

| $v$ | 2 | 4 | 10 |
|---|---|---|---|
| $A$ | 12 | 48 | 300 |

..... **[3]**

**16** **(a)** The table shows some values of $x$ and $y$.

| $x$ | 2 | 4 | 5 | 8 |
|---|---|---|---|---|
| $y$ | 32 | 8 | 5.12 | 2 |

**(i)** Identify the variation model and write down the equation for $y$ in terms of $x$.

$y =$ ..... **[3]**

**(ii)** Calculate the value of $y$ when $x = 10$.

$y =$ ..... **[1]**

**(b)** $A$ is directly proportional to the cube of $B$.
When $B$ is increased by $x\%$, the value of $A$ increases by 72.8%.
Find the value of $x$.

$x =$ ..... **[3]**

**17** **(a)** $y$ is inversely proportional to the square root of $x$.

When $x$ is increased by 44%, find the percentage change in $y$.

..... % **[4]**

**(b)** $P$ is directly proportional to the cube of $Q$.
$Q$ is inversely proportional to $R$.
When $R = 2$, $P = 10$.

Find the value of $P$ when $R = 4$.

$P =$ ..... **[3]**

**18** The table shows some corresponding values of $x$ and $y$.

| $x$ | 2 | 4 | 10 |
|---|---|---|---|
| $y$ | 40 | 5 | 0.32 |

**(a)** Determine the proportional relationship between $y$ and $x$ and find an equation connecting them.

$y =$ ..... **[3]**

**(b)** Find the value of $x$ when $y = 2560$.

$x =$ ..... **[2]**

**(c)** The variable $x$ is inversely proportional to $w$.
When $w = 5$, $x = 2$.

Find an equation connecting $y$ and $w$.
Give your answer in its simplest form.

$y =$ ..... **[4]**

**1** **(a)** $y = kx^2$ or $36 = k(3^2)$ **[M1]**

$4x^2$ **[A1]**

**(b)** 100 **[B1]**

[Back to Question] *[Total Marks: 3]*

**2** $y = k(x + 2)$ **[M1]**

$20 = k(3 + 2)$ leading to $k = 4$ **[M1]**

36 **[A1]**

[Back to Question] *[Total Marks: 3]*

**3** **(a)** Sets up equation $y = kx$ and substitutes $x = 5, y = 15$ to get $15 = 5k$ **[M1]**

$y = 3x$ **[A1]**

**(b)** Substitutes $x = 12$ into their equation from part (a), e.g. $y = 3 \times 12$ **[M1]**

36 **[A1]**

[Back to Question] *[Total Marks: 4]*

**4** **(a)** Substitutes into the proportion equation to find the constant, e.g. $135 = k(1 + 2)^3$ **[M1]**

$5(x + 2)^3$ **[A1]**

**(b)** 40 **[B1]**

**(c)** Sets up the equation $320 = 5(x + 2)^3$ and simplifies to $(x + 2)^3 = 64$ **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 5]*

**5** **(a)** $y = kx^2$ leading to $36 = k(3)^2$ **[M1]**

$4x^2$ **[A1]**

**(b)** 100 **[B1]**

**(c)** $196 = 4x^2$ leading to $x^2 = 49$ **[M1]**

7 and $-7$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**6** **(a)** $y = kx^3$ **[M1]**

$24 = k(2)^3 \implies k = 3$ **[M1]**

$3x^3$ **[A1]**

**(b)** $y = 3(5)^3 = 3(125)$ **[M1]**

375 **[A1]**

**(c)** $81 = 3x^3$ **[M1]**

$x^3 = 27$ **[M1]**

3 **[A1]**

[Back to Question] *[Total Marks: 5]*

**7** **(a)** Recognises inverse square proportion, e.g. tests $p^2q$ **[M1]**

$2^2 \times 100 = 400$, $4^2 \times 25 = 400 \implies k = 400$ **[M1]**

$\dfrac{400}{p^2}$ **[A1]**

**(b)** $q = \dfrac{400}{5^2} = \dfrac{400}{25}$ **[M1]**

16 **[A1]**

**(c)** $1 = \dfrac{400}{p^2}$ **[M1]**

$p^2 = 400$ **[M1]**

20 (Accept $\pm 20$) **[A1]**

[Back to Question] *[Total Marks: 5]*

**8** **(a)** Sets up $P = k_1\sqrt{Q}$ and substitutes
$Q = 9, P = 12 \implies 12 = k_1\sqrt{9} \implies 12 = 3k_1 \implies k_1 = 4$, so $P = 4\sqrt{Q}$ **[M1]**

Sets up $Q = \dfrac{k_2}{R^3}$ and substitutes $R = 2, Q = 2 \implies 2 = \dfrac{k_2}{2^3} \implies 2 = \dfrac{k_2}{8} \implies k_2 = 16$, so
$Q = \dfrac{16}{R^3}$ **[M1]**

Substitutes $Q$ into the $P$ equation: $P = 4\sqrt{\dfrac{16}{R^3}}$ **[M1]**

Simplifies the square root: $P = 4 \times \dfrac{4}{\sqrt{R^3}}$ **[M1]**

$P = \dfrac{16}{R^{1.5}}$ or $P = 16R^{-1.5}$ or $P = \dfrac{16}{R\sqrt{R}}$ **[A1]**

**(b)** Substitutes $R = 4$ into their formula: $P = \dfrac{16}{4\sqrt{4}}$ or $P = \dfrac{16}{8}$ **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 7]*

**9** **(a)** Sets up equation $y = \dfrac{k}{x^2}$ **[M1]**

Substitutes $x = 4, y = 3$ to get $3 = \dfrac{k}{4^2} \implies k = 3 \times 16 = 48$ **[M1]**

$y = \dfrac{48}{x^2}$ **[A1]**

**(b)** Sets $0.75 = \frac{48}{x^2}$ **[M1]**

Rearranges to $x^2 = \frac{48}{0.75} = 64$ **[M1]**

$x = 8, x = -8$ (both required for final mark) **[A1]**

[Back to Question] *[Total Marks: 6]*

**10 (a)** $P = \frac{k}{Q^2}$ **[M1]**

$1.5 = \frac{k}{4^2}$ **[M1]**

$\frac{24}{Q^2}$ **[A1]**

**(b)** 0.24 or $\frac{6}{25}$ **[B1]**

**(c)** $24 = \frac{24}{Q^2}$ or $Q^2 = 1$ **[M1]**

1 **[A1]**

[Back to Question] *[Total Marks: 6]*

**11 (a) (i)** $P = \frac{k}{\sqrt{Q}}$ leading to $12 = \frac{k}{\sqrt{9}}$ **[M1]**

$k = 36$ **[B1]**

3 **[A1]**

**(ii)** $1.5 = \frac{36}{\sqrt{Q}}$ leading to $\sqrt{Q} = 24$ **[M1]**

576 **[A1]**

**(b)** Identifying that the new value of $P$ involves dividing by $\sqrt{4}$ or 2 **[M1]**

50 **[A1]**

[Back to Question] *[Total Marks: 7]*

**12 (a)** $M = \frac{k}{\sqrt{N}}$ or $15 = \frac{k}{\sqrt{16}}$ **[M1]**

$\frac{60}{\sqrt{N}}$ **[A1]**

**(b)** $\frac{60}{\sqrt{100}}$ or $\frac{60}{10}$ **[M1]**

6 **[A1]**

**(c)** $1.2 = \frac{60}{\sqrt{N}}$ **[M1]**

$\sqrt{N} = \frac{60}{1.2}$ or $\sqrt{N} = 50$ **[M1]**

2500 **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**13** **(a)** Substitutes into the proportion equation to find the constant, e.g. $4 = \frac{k}{\sqrt{36}}$ **[M1]**

$\frac{24}{\sqrt{Q}}$ **[A1]**

**(b)** 2 **[B1]**

**(c)** Sets up the equation $1.5 = \frac{24}{\sqrt{Q}}$ leading to $\sqrt{Q} = 16$ **[M1]**

256 **[A1]**

**(d)** Identifies the new value of $Q$ is $4Q$, or uses numerical values (e.g. $Q$ increases from 36 to 144) **[M1]**

Evaluates the new value of $P$ as $\frac{1}{2}P$, or calculates the new $P$ is 2 when original was 4 **[M1]**

50 **[A1]**

**[Back to Question]** *[Total Marks: 8]*

**14** **(a)** Tests products or ratios, e.g., shows $x^2y = 300$ for all pairs or identifies $k = 300$ **[M1]**

Identifies $y \propto \frac{1}{x^2}$ as the correct model **[A1]**

$\frac{300}{x^2}$ **[A1]**

**(b)** Uses 1.5 multiplier and finds $(1.5)^3$ or 3.375 **[M1]**

Identifies increase factor as $3.375 - 1$ or 2.375 **[M1]**

237.5 **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**15** **(a)** Uses new $x = 1.2x$ to write $y_{new} = k(1.2x)^3$ **[M1]**

Calculates $1.2^3 = 1.728$ **[M1]**

72.8 **[A1]**

**(b)** Tests one incorrect model and shows the constant is not equal, e.g. for $A \propto v$, calculates $\frac{12}{2} = 6$ but $\frac{48}{4} = 12$ **[M1]**

Tests $A \propto v^2$ and shows $\frac{A}{v^2}$ is constant by calculating $\frac{12}{4} = 3$, $\frac{48}{16} = 3$, and $\frac{300}{100} = 3$ **[M1]**

$A \propto v^2$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**16** **(a)** **(i)** Recognises that $x^2y$ is constant (128), or identifies the model as $y \propto \frac{1}{x^2}$ **[M1]**

Sets up the equation $y = \frac{k}{x^2}$ and correctly calculates $k = 128$ **[M1]**

$\frac{128}{x^2}$ **[A1]**

**(ii)** 1.28 **[B1]**

**(b)** Sets up the correct scale factor equation for $B$, e.g. $m^3 = 1.728$ or $\left(1 + \frac{x}{100}\right)^3 = 1.728$ **[M1]**

Calculates the multiplier $m = \sqrt[3]{1.728} = 1.2$ **[M1]**

20 **[A1]**

[Back to Question] *[Total Marks: 7]*

**17** **(a)** Forms equation: $y = \frac{k}{\sqrt{x}}$ **[M1]**

Uses multiplier 1.44 for the increase in $x$: New $y = \frac{k}{\sqrt{1.44x}}$ **[M1]**

Simplifies denominator: $\sqrt{1.44} = 1.2$, so New $y = \frac{1}{1.2} \times \frac{k}{\sqrt{x}}$ (or $\frac{5}{6}y$) **[M1]**

Calculates percentage change: $\left(\frac{5}{6} - 1\right) \times 100$ **[M1]**

$-16.7$ (or 16.7 decrease) **[A1]**

**(b)** Deduces relationship between $P$ and $R$: $P = k_1\left(\frac{k_2}{R}\right)^3 = \frac{K}{R^3}$ (or $P \propto \frac{1}{R^3}$) **[M1]**

Substitutes given values to find constant: $10 = \frac{K}{2^3} \implies K = 80$ **[M1]**

Substitutes $R = 4$ into their equation: $P = \frac{80}{4^3} = \frac{80}{64}$ **[M1]**

1.25 (or $\frac{5}{4}$) **[A1]**

[Back to Question] *[Total Marks: 7]*

**18** **(a)** Demonstrating that $y \times x^3$ is constant for at least two pairs (e.g., $40 \times 2^3 = 320$ and $5 \times 4^3 = 320$) **[M1]**

$y \propto \frac{1}{x^3}$ **[B1]**

$\frac{320}{x^3}$ **[A1]**

**(b)** $2560 = \frac{320}{x^3}$ leading to $x^3 = \frac{1}{8}$ **[M1]**

$\frac{1}{2}$ or 0.5 **[A1]**

**(c)** $x = \frac{k_2}{w}$ leading to $2 = \frac{k_2}{5}$ **[M1]**

$x = \dfrac{10}{w}$ **[A1]**

Substituting into their equation for $y$: $y = \dfrac{320}{(\frac{10}{w})^3}$ **[M1]**

$0.32w^3$ or $\dfrac{8}{25}w^3$ **[A1]**

**[Back to Question]** *[Total Marks: 9]*